Stochastic programming
Multistage problems

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Motivation: financial planning

(Birge and Louveaux, Section 1.2)

- We invest an initial wealth b, and would like to reach a goal G after Y years.
- The portfolio is rebalanced every v years: there are $H = Y/v$ investment periods, $\mathcal{T} = \{1, \ldots, H\}$, and an investment decision at the beginning of each period.
- N: set of available assets (stocks, bonds, ...).
- If the final wealth exceeds the goal G, the surplus earns interest at rate q.
- If it falls short of G, the shortfall must be borrowed at rate r (typically, $r > q$).

Variables and data

- x_{it}, $i \in N$, $t \in \mathcal{T}$: amount invested in asset i at the beginning of period t;
- ω_{it}: return of asset i during period t, i.e., the value at the end of the period of one dollar invested at its beginning;
- y: surplus over the goal at the end of the horizon;
- w: shortfall at the end of the horizon.

When the returns are random, $x_{i2}, \ldots, x_{iH}$ are decided after observing the returns of the previous periods: they rebalance the portfolio, and are recourse actions toward the goal G.

Deterministic formulation

$$
\begin{aligned}
\max\ & qy - rw \\
\text{s.t.}\ & \sum_{i \in N} x_{i1} = b, \\
& \sum_{i \in N} \omega_{i,t-1} x_{i,t-1} = \sum_{i \in N} x_{it}, \quad t = 2, \ldots, H, \\
& \sum_{i \in N} \omega_{iH} x_{iH} - y + w = G, \\
& x_{it} \geq 0, \quad i \in N,\ t \in \mathcal{T}, \qquad y, w \geq 0.
\end{aligned}
$$

The wealth at the end of period $t-1$ is entirely reinvested at the beginning of period t.

Random returns

- The returns are not known in advance: they are random variables.
- Assume that, in each period t, the vector of returns $(\omega_{it})_{i \in N}$ takes one of R possible values.
- A scenario is a sequence of realizations, one per period: there are R^H scenarios.

Example: $R = 4$ and $H = 2$, hence 16 scenarios (one per row; each entry is the index of the realization in that period):

$t = 1$	$t = 2$
1	1
1	2
1	3
1	4
2	1
⋮	⋮
4	4

Stochastic program formulation

- S: set of scenarios; p_s: probability of scenario $s \in S$;
- ω_{its}: return of asset i during period t in scenario s;
- x_{its}: amount invested in asset i at the beginning of period t in scenario s;
- y_s, w_s: surplus and shortfall at the end of the horizon in scenario s.

Each variable is copied once per scenario, and we maximize the expected value of $qy - rw$.

Stochastic version

$$\max \sum_{s\in S} p_s(qy_s - rw_s)$$

$$\text{s.t.} \sum_{i\in N} x_{i1s} = b, \quad \forall s \in S,$$

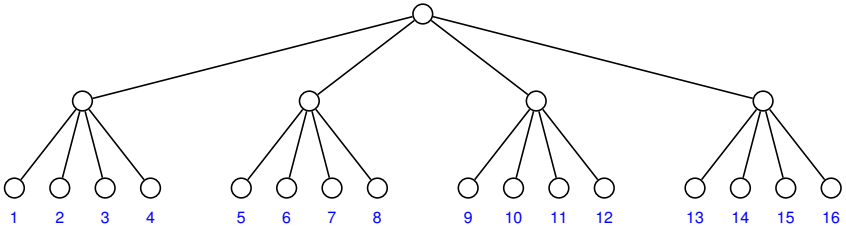
$$\sum_{i\in N} \omega_{i,t-1,s}\, x_{i,t-1,s} = \sum_{i\in N} x_{its}, \quad t = 2,\dots,H, \ \forall s \in S,$$

$$\sum_{i\in N} \omega_{iHs}\, x_{iHs} - y_s + w_s = G, \quad \forall s \in S,$$

$$x_{its} \geq 0, \quad i \in N, \ t \in \mathcal{T}, \ s \in S, \qquad y_s, w_s \geq 0, \quad s \in S.$$

Scenario tree

Conceptually, the sequence of events can be arranged in a tree: the root is the first decision, each branching is the realization of the returns of a period, and each leaf is a scenario. Example ($R = 4$, $H = 2$):



The scenarios can have different probabilities, and the returns of successive periods can be dependent: the branching probabilities can differ from one node to another.

Nonanticipativity

- Converting the deterministic model into the stochastic version is easy...
- but the resulting model is wrong: x_{its} may depend on the whole scenario s, including the returns of periods $t, \ldots, H$, unknown when x_{its} is decided. The model sees the future.
- We have to enforce nonanticipativity: a decision may only depend on the information available when it is made.

Let S_s^t be the set of scenarios identical to s up to period $t-1$, i.e., indistinguishable from s when the decisions of period t are made ($S_s^1 = S$). We impose

$$x_{its} = x_{its'}, \quad \forall i \in N,\ t \in \mathcal{T},\ s \in S,\ s' \in S_s^t.$$

The final y_s and w_s are decided once everything is known: they need no such constraint.

Nonanticipativity (cont'd)

Equivalently, as every $p_s > 0$,

$$\Big(\sum_{s' \in S_s^t} p_{s'}\Big) x_{its} = \sum_{s' \in S_s^t} p_{s'} x_{its'}, \quad \forall i, t, s,$$

that is,

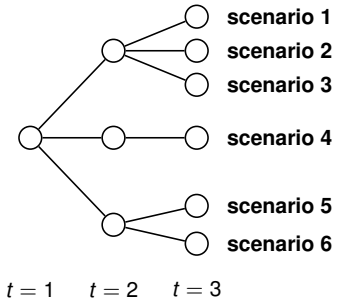
$$x_{its} = \frac{\sum_{s' \in S_s^t} p_{s'} x_{its'}}{\sum_{s' \in S_s^t} p_{s'}} = \mathbb{E}\left[x_{its'} \mid s' \in S_s^t\right], \quad \forall i, t, s.$$

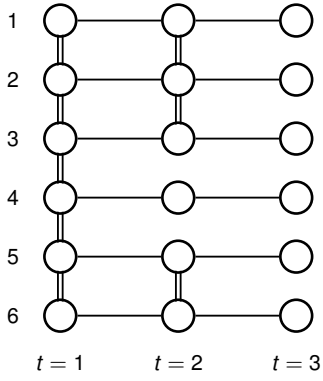
Each decision equals its conditional expectation given the information available when it is made: if all the decisions of a group of indistinguishable scenarios equal the group average, they are all equal. This form underlies progressive hedging (Lagrangian methods slides).

There are many other formulations, for instance equalities between consecutive scenarios of each group (next slide).

Back to the scenario tree

In the scenario formulation, each decision is copied once per scenario, and nonanticipativity ties the copies together.





Left: six scenarios, three periods. Right: one row per scenario, one column per period; a double bar is an equality $x_{its} = x_{its'}$.

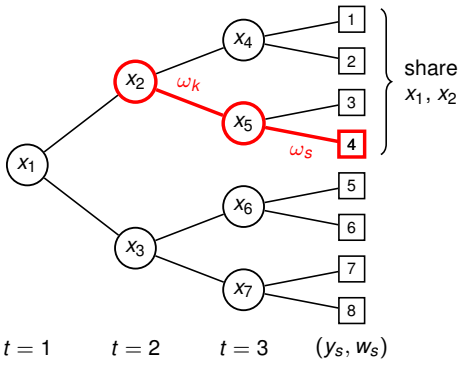
Nodal approach

- Alternatively, define one vector of decisions per node of the tree: the decision made given the realizations observed up to this node.
- Nonanticipativity is then built in: all the scenarios going through a node share its decision.
- $\mathcal{L}$: set of the nodes where investments are decided, with root 1; the leaves are the scenarios $s \in S$, where y_s and w_s are decided.
- $a(k)$: parent of node k (or of leaf s), i.e., its ancestor at the previous stage; ω_{ik}: return of asset i on the branch from $a(k)$ to k.
- Fewer variables, $|N|\,|\mathcal{L}|$ instead of $|N|\,H\,|S|$, and no nonanticipativity constraint.

Nodal formulation

$$
\begin{aligned}
\max \quad & \sum_{s \in S} p_s (q y_s - r w_s) \\
\text{s.t.} \quad & \sum_{i \in N} x_{i1} = b, \\
& \sum_{i \in N} \omega_{ik}\, x_{i,a(k)} = \sum_{i \in N} x_{ik}, \quad \forall k \in \mathcal{L} \setminus \{1\}, \\
& \sum_{i \in N} \omega_{is}\, x_{i,a(s)} - y_s + w_s = G, \quad \forall s \in S, \\
& x_{ik} \geq 0, \quad i \in N,\ k \in \mathcal{L}, \qquad y_s, w_s \geq 0, \quad s \in S.
\end{aligned}
$$

Nodal formulation on the scenario tree



Financial example, $H = 3$ periods, two outcomes per period.

- One vector x_k per node $k \in \mathcal{L}$ (7 nodes), one pair (y_s, w_s) per leaf $s \in S$ (8 scenarios).
- Node $k = 5$ reinvests the wealth of its parent $a(k) = 2$: $\sum_i \omega_{ik}\, x_{i,a(k)} = \sum_i x_{ik}$.
- Leaf $s = 4$ checks the goal: $\sum_i \omega_{is}\, x_{i,a(s)} - y_s + w_s = G$.
- Scenarios 1 to 4 share x_1 and x_2: nonanticipativity is built in.
- $7\,|N|$ investment variables, against $H\,|S|\,|N| = 24\,|N|$ in the scenario formulation.

Another example: multi-period production planning

- A firm produces several goods over H periods, with limited resources (labor, machine hours), extra resources that can be bought, and a cost for changing the workforce.
- The demand of each period is random and must be met: missing products are bought from an external source, unsold ones are stored, both at a cost.
- Here and now: the production plan of every period, the extra resources, the hiring and firing. Recourse: in each period, once its demand is known, how much to store and how much to buy; these decisions may only use the demands observed so far.
- Nonanticipativity only constrains the recourse if buying ahead can pay, for instance when purchase prices rise faster than storage costs. Otherwise, buying the deficit is optimal whatever the future demands, and the problem behaves like a two-stage problem.

The model, these two cases, and its writing as a multistage linear program are an exercise (exercise collection, section “Multistage problems”).

Multistage linear programs

A stage is a decision time. Decisions and observations alternate,

$$x_1 \;\to\; \xi_2 \;\to\; x_2 \;\to\; \xi_3 \;\to\; \cdots \;\to\; \xi_H \;\to\; x_H,$$

and x_t may only depend on the history $\xi_{[t]} = (\xi_1, \ldots, \xi_t)$, ξ_1 being deterministic. Over the horizon $t = 1, \ldots, H$, the multistage stochastic linear program is

$$\min_{x_1 \geq 0} \; c_1^T x_1 + \mathbb{E}\Big[\min_{x_2 \geq 0} c_2^T x_2 + \mathbb{E}\big[\cdots + \mathbb{E}[\min_{x_H \geq 0} c_H^T x_H]\big]\Big]$$

subject to $W_1 x_1 = h_1$ and $W_t x_t = h_t(\xi_t) - T_{t-1}(\xi_t) x_{t-1}$, $t = 2, \ldots, H$ (c_t may also depend on ξ_t).

- Each expectation is conditional on the history, and each min over x_t is taken once ξ_t is observed, before ξ_{t+1}: the nesting enforces nonanticipativity.
- For $H = 2$, this is the two-stage recourse problem, with $A = W_1$, $b = h_1$, $y = x_2$, $q = c_2$, $W = W_2$, $h = h_2$, $T = T_1$.

Financial planning as a multistage program

Stages $t = 1, \ldots, H$: the investments $x_t = (x_{it})_{i \in N}$; stage $H+1$: $x_{H+1} = (y, w)$. Between stages $t-1$ and t, the returns of period $t-1$ are observed: $\xi_t = (\omega_{i,t-1})_{i \in N}$. With $\mathbf{1}$ the vector of ones (maximizing $qy - rw$ is minimizing $-qy + rw$):

stage	c_t	W_t	h_t	$T_{t-1}(\xi_t)$
$t = 1$	0	$\mathbf{1}^T$	b	–
$t = 2, \ldots, H$	0	$\mathbf{1}^T$	0	$-\xi_t^T$
$t = H+1$	$(-q, r)$	$(-1, 1)$	G	ξ_{H+1}^T

For instance, $W_t x_t = h_t - T_{t-1} x_{t-1}$ reads $\mathbf{1}^T x_t = \xi_t^T x_{t-1}$ for $2 \leq t \leq H$: the wealth is reinvested.

- H periods, but $H+1$ stages: the surplus and the shortfall are a last recourse decision, made once all the returns are known.
- The randomness lies in the technology matrices only.

Potential solution techniques

Adapted from Jalal Kazempour,
`https://www.jalalkazempour.com/teaching`

- Direct solution: solve the extensive form with a linear programming solver; its size grows exponentially with the number of stages.
- Linear decision rules: restrict each decision to an affine function of the uncertain parameters observed so far, which gives an upper bound, for a minimization. If the support of ξ is a polytope, and the constraints must hold for every ξ in it, duality yields a linear program whose size grows only polynomially with the number of stages (Kuhn, Wiesemann and Georghiou, 2011).
- Nested decomposition (nested Benders): fixes the decisions of the earlier stages (complicating variables), and decomposes the problem by node of the tree.
- Lagrangian relaxation: relaxes the nonanticipativity constraints (complicating constraints), and decomposes the problem by scenario.

Dynamic programming recursion

Cost-to-go functions, with $\mathcal{Q}_{H+1} \equiv 0$:

$$V_t(x_{t-1}, \xi_{[t]}) = \min_{x_t \geq 0} \left\{ c_t^T x_t + \mathcal{Q}_{t+1}(x_t, \xi_{[t]}) \ : \ W_t x_t = h_t(\xi_t) - T_{t-1}(\xi_t) x_{t-1} \right\},$$
$$\mathcal{Q}_{t+1}(x_t, \xi_{[t]}) = \mathbb{E}\left[V_{t+1}(x_t, \xi_{[t+1]}) \mid \xi_{[t]} \right].$$

- V_t: optimal cost of stages $t, \ldots, H$, given the previous decision and the history; $\mathcal{Q}_{t+1}$: the expected cost-to-go, its expectation given the history.
- The problem is $\min\{c_1^T x_1 + \mathcal{Q}_2(x_1, \xi_1) : W_1 x_1 = h_1, \ x_1 \geq 0\}$: a two-stage problem whose recourse function is itself defined by a multistage problem.
- On a finite tree, there is one function $\mathcal{Q}_{t+1}(\cdot, \xi_{[t]})$ per node of stage t, convex and piecewise linear on its domain.

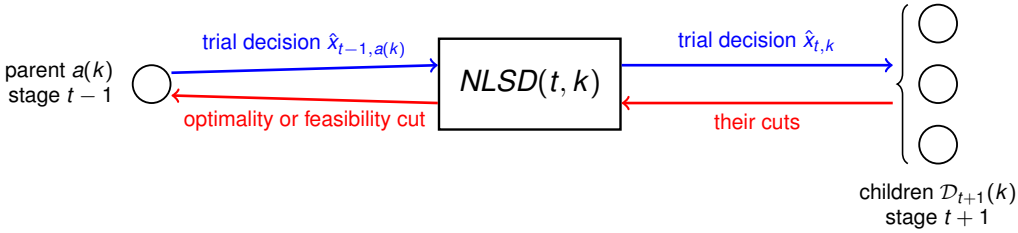
Nested decomposition

The nested L-shaped method (Birge, 1985) applies the L-shaped method recursively on the tree. Node k at stage t, master of its children and subproblem of its parent $a(k)$, solves the nested L-shaped decomposition subproblem $NLSD(t,k)$:

$$\begin{aligned}
\min_{x_{t,k}\geq 0,\,\theta_{t,k}} \; & c_{t,k}^T x_{t,k} + \theta_{t,k} \\
\text{s.t.} \; & W_t x_{t,k} = h_{t,k} - T_{t-1,k}\,\hat{x}_{t-1,a(k)}, \quad (\pi_{t,k}) \\
& D_{t,k,\ell}\, x_{t,k} \geq d_{t,k,\ell},\ \ell = 1,\ldots,r_{t,k}, \quad (\rho_{t,k,\ell}) \quad \text{feasibility cuts,} \\
& E_{t,k,\ell}\, x_{t,k} + \theta_{t,k} \geq e_{t,k,\ell},\ \ell = 1,\ldots,s_{t,k}, \quad (\sigma_{t,k,\ell}) \quad \text{optimality cuts.}
\end{aligned}$$

- $c_{t,k}$, $h_{t,k}$, $T_{t-1,k}$: values of c_t, h_t, T_{t-1} at node k; $\hat{x}_{t-1,a(k)}$: trial decision of the parent; $r_{t,k}$, $s_{t,k}$: numbers of cuts received so far;
- $\theta_{t,k}$ underestimates the expected cost-to-go at node k, $\mathcal{Q}_{t+1}(x_{t,k}, \xi_{[t]})$, and is bounded below by a constant, like θ^0 in the L-shaped dual decomposition; $\theta_{H,k} \equiv 0$.

Nested decomposition: what a node exchanges



- Subproblem of its parent (left): solved at the trial decision of $a(k)$, it returns a cut built from its optimal value $v_{t,k}$ and the multipliers $\pi_{t,k}$, or a feasibility cut if it is infeasible (next slides).
- Master problem of its children (right): its solution $\hat{x}_{t,k}$ is their trial decision, and their cuts become the rows $D_{t,k,\ell}$ and $E_{t,k,\ell}$ of $NLSD(t,k)$.
- Blue: forward pass, from the root to the leaves; red: backward pass, back to the root. The root has no parent, a leaf no child.

Nested decomposition: cuts

Let $v_{t,k}$ be the optimal value of node k, and $p(k, t \mid j, t-1)$ the probability of moving from node j to its child $k \in \mathcal{D}_t(j)$. The optimality cut of node j at stage $t-1$, a new row of $NLSD(t-1, j)$, averages the supporting hyperplanes of its children:

$$\theta_{t-1,j} \geq \sum_{k \in \mathcal{D}_t(j)} p(k, t \mid j, t-1) \left(v_{t,k} - \pi_{t,k}^T T_{t-1,k} \left(x_{t-1,j} - \hat{x}_{t-1,j} \right) \right).$$

- Why the cut is valid: with its current cuts, node k defines a convex function of x_{t-1}, below $V_t(\cdot, \xi_{[t]})$ since $\theta_{t,k}$ underestimates. By linear programming duality, $v_{t,k} - \pi_{t,k}^T T_{t-1,k}(x_{t-1} - \hat{x}_{t-1,j})$ supports this function at $\hat{x}_{t-1,j}$. Averaging over the children gives an affine function below $\mathcal{Q}_t(\cdot, \xi_{[t-1]})$.
- The slope only involves the multipliers $\pi_{t,k}$ of the constraints linking node k to its parent; the cuts of node k enter through $v_{t,k}$: information propagates up the tree.

Nested decomposition: feasibility cuts

If $NLSD(t,k)$ is infeasible for the trial decision $\hat{x}_{t-1,j}$ of its parent $j = a(k)$, the solver returns a certificate of infeasibility (Farkas' lemma, linear programming background notes): $\pi_{t,k}$ and $\rho_{t,k} \geq 0$ such that

$$\pi_{t,k}^T W_t + \rho_{t,k}^T D_{t,k} \leq 0 \quad \text{and} \quad \pi_{t,k}^T \left(h_{t,k} - T_{t-1,k}\, \hat{x}_{t-1,j} \right) + \rho_{t,k}^T d_{t,k} > 0,$$

where $D_{t,k}$ and $d_{t,k}$ stack the feasibility cuts of node k ($\theta_{t,k}$ plays no role).

- If $x_{t,k} \geq 0$ solves node k for some x_{t-1}, then $\pi_{t,k}^T (h_{t,k} - T_{t-1,k} x_{t-1}) + \rho_{t,k}^T d_{t,k} \leq (\pi_{t,k}^T W_t + \rho_{t,k}^T D_{t,k})\, x_{t,k} \leq 0$. Hence the feasibility cut, a new row of $NLSD(t-1,j)$,

  $$\pi_{t,k}^T T_{t-1,k}\, x_{t-1,j} \geq \pi_{t,k}^T h_{t,k} + \rho_{t,k}^T d_{t,k},$$

  satisfied by every decision that keeps node k feasible, and violated by $\hat{x}_{t-1,j}$.
- If the parent becomes infeasible, it sends a feasibility cut to its own parent; if the first stage does, the problem is infeasible. None is needed with relatively complete recourse, as in the financial example.

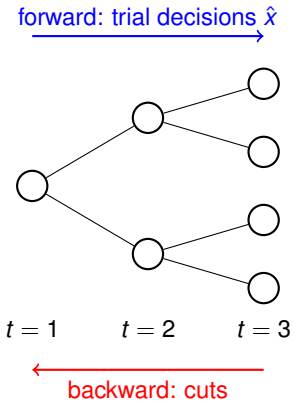
Nested decomposition: forward and backward passes

Forward pass: solve the first-stage problem, send $\hat{x}_1$ to its children, solve them, and so on down to stage H. An infeasible node sends a feasibility cut to its parent, and a new forward pass starts. Otherwise, the trial decisions form a nonanticipative policy, whose expected cost is an upper bound.

Backward pass: for $t = H, \ldots, 2$, the nodes of stage t send optimality cuts to their parents, solved again with these cuts. The first-stage value is a lower bound.

Stop when the bounds meet. On a finite tree, with bounded x_t, only finitely many distinct cuts exist: finite convergence (Birge and Louveaux, Section 6.1, Theorem 1). Other sequencing protocols change direction before reaching the last or the first stage.

Julia code: `code/nested_benders.ipynb`.



Beyond nested decomposition

The scenario tree grows exponentially with the number of stages, and so does the number of problems solved at each iteration of nested decomposition.

- Stochastic dual dynamic programming (Pereira and Pinto, 1991; SDDP slides): under stagewise independence, all the nodes of a stage share the same expected cost-to-go function, hence the same cuts, and the forward pass samples scenarios instead of enumerating the tree. Extensions handle Markovian processes, on a lattice.
- Lagrangian methods (progressive hedging; Lagrangian methods slides): relax the nonanticipativity constraints, and decompose the problem by scenario, each subproblem being a deterministic multistage problem. They do not require a Markovian process, but there is one subproblem per scenario, and many iterations can be needed to restore nonanticipativity.